Trainings

Preparatory Steps

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Use skin and eye protection when handling caustic solutions to reduce the possibility of personal injury.

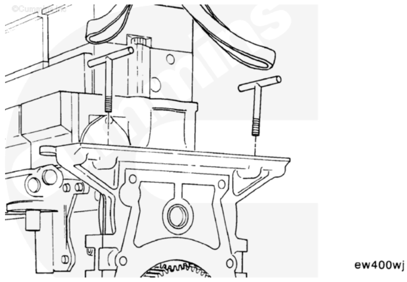
- Remove the upper rear gear drive assembly. Refer to Procedure 009-024 (/qs3/pubsys2/xml/en/procedures/20/20-009-024-tr.html).
- Remove the starter. Refer to Procedure 013-020 (/qs3/pubsys2/xml/en/procedures/20/20-013-020-tr.html).
- Remove the transmission, clutch, and all related components. Refer to the equipment manufacturer's instructions.
- Remove the flywheel. Refer to Procedure 016-005 (/qs3/pubsys2/xml/en/procedures/20/20-016-005-tr.html).
- Drain the lubricating oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/20/20-007-037.html).
- Remove the crankshaft rear seal. Refer to Procedure 001-024 (/qs3/pubsys2/xml/en/procedures/20/20-001-024-tr.html).
- Remove the oil pan adapter cover plate. Refer to Procedure 007-026 (/qs3/pubsys2/xml/en/procedures/20/20-007-026-tr.html).
- Remove the flywheel housing. Refer to Procedure 016-006 (/qs3/pubsys2/xml/en/procedures/20/20-016-006-

tr.html).

Remove

⚠ WARNING ⚠

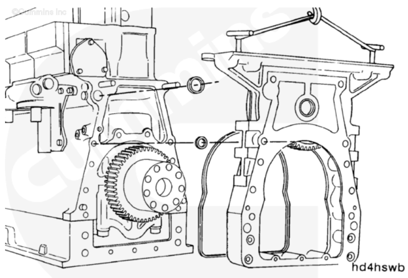
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a hoist, two tee handles, and a lifting sling. Install the tee handles. Adjust the hoist until there is tension in the lifting sling.

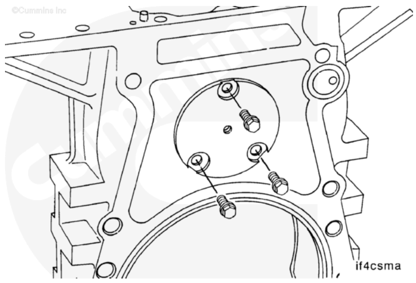
Use a mallet to tap the lower housing of the rear gear train off the two locating dowel pins in the rear face of the cylinder block.

Remove and discard the rectangular seal, bolt seals, and main oil rifle seal.

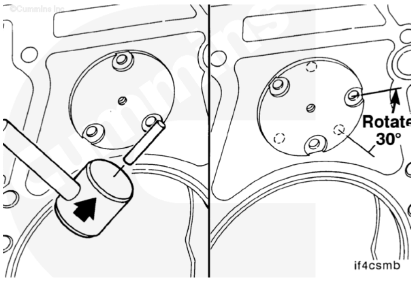


Disassemble

Remove the three capscrews from the idler shaft



If the idler has three holes, use a soft punch and mallet. Rotate the idler shaft 30 degrees until the capscrew holes in the housing are **not** visible through the capscrew holes in the idler shaft.

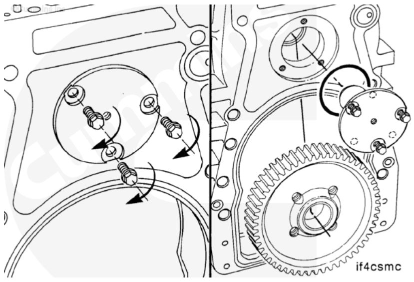


Use three 3/8-24x2 inch capscrews. Make sure the idler gear is secure.

Alternately tighten the capscrews to pull the shaft **from** the housing.

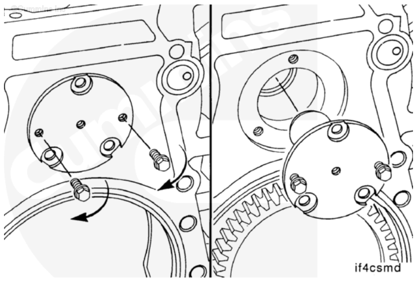
Remove the shaft, thrust washers, and gear.

Remove and discard the o-ring.



If the idler shaft has five holes, use two 3/8-24x2 inch capscrews. Make sure the idler gear is secure.

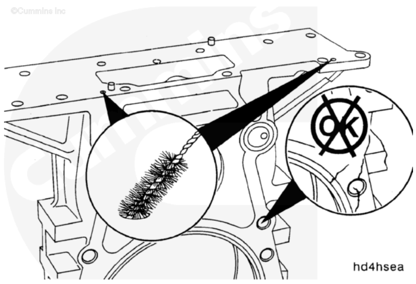
Alternately tighten the capscrews to pull the shaft **from** the housing.



Inspect for Reuse

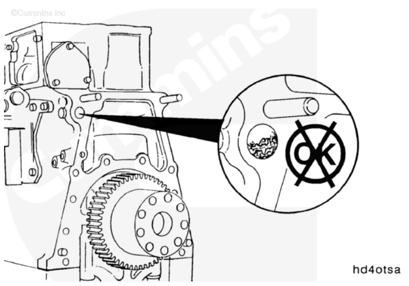
Remove all pipe plugs and clean all oil drillings. Be sure the oil drillings intersect.

Clean and check the lower housing for damage.



⚠ CAUTION ⚠

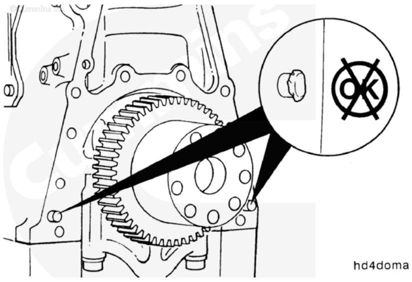
Check for debris in the main oil rifle. Any debris will cause rear gear train damage.



Be sure the 1-1/8 inch expansion plug is **not** installed in the end of the main oil rifle in the cylinder block. The main oil rifle **must** be OPEN to allow the oil to the rear gear train.

Inspect the two rear gear train housing dowel pins in the rear face of the cylinder block. Any damaged dowels **must** be replaced.

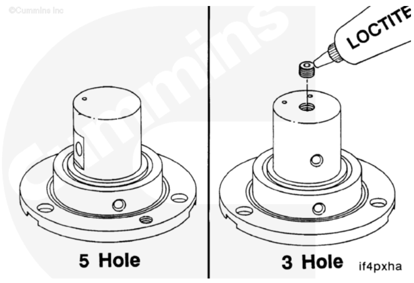
Use dowel pin extractor, Part Number ST-1143, to remove the dowel pins. Use a brass or lead hammer to install the new dowel pins.



Assemble

⚠ CAUTION ⚠

Absence of the pipe plug can cause serious engine damage due to a loss in oil pressure to the rear gear train bushings.



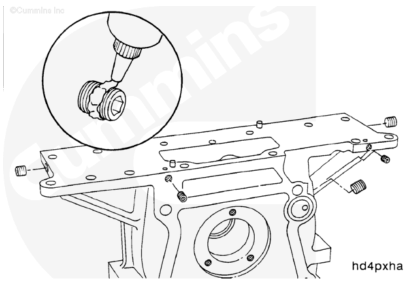
The five hole style idler shaft does **not** have a hole for the internal hex pipe plug.

Use Loctite™ to install the internal hex pipe plug into the end of the idler shaft.

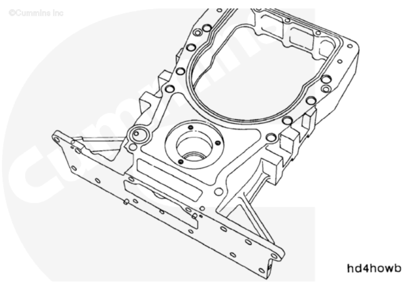
Torque Value: 11 n•m [97 in-lb]

Use pipe sealant, Part Number 3375066, or equivalent sealant with liquid Teflon®. Install five 1/8 inch internal hex pipe plugs into the lower housing.

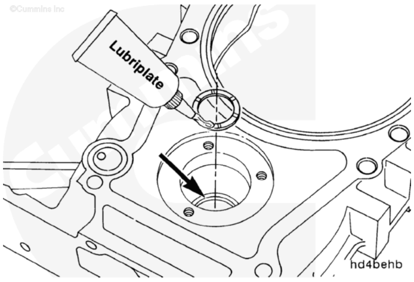
Torque Value: 16 n•m [142 in-lb]



Position the lower housing horizontally with the block mating surface facing up. Attempting to install the gear and idler shaft with the housing vertical can result in binding of the thrust bearings.

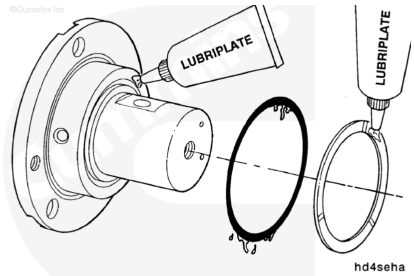


Use Lubriplate® 105, multi-purpose lubricant, or equivalent. Lubricate one thrust bearing and the counterbore in the lower housing near the expansion plug. Install the thrust bearing into the counterbore with the grooved side facing **away** from the housing.



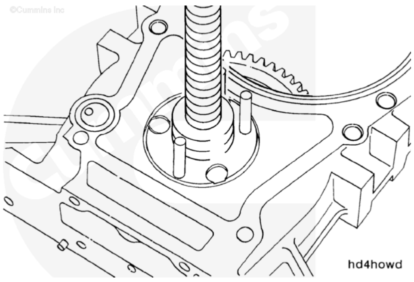
Install 5/16-18x2 inch guide studs.

Align the idler gear into the lower housing. Align and install the idler shaft into the idler gear and lower housing until the idler shaft pilot chamfer touches the lower housing.



Make sure the machined surfaces on the back side of the lower housing are supported to prevent cracking of the housing.

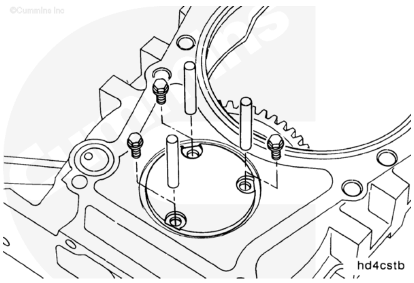
Use an arbor press to press the idler shaft into the lower housing until the mounting flange contacts the housing.



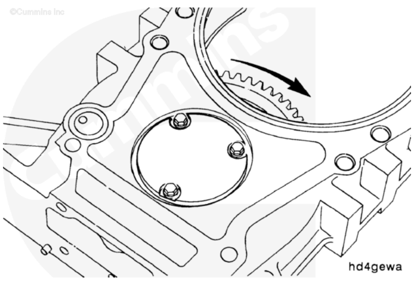
Remove the guide studs and install the three 5/16-18x3/4 inch capscrews.

Tighten the three capscrews.

Torque Value: 20 n•m [177 in-lb]



Rotate the gear to check for binding.

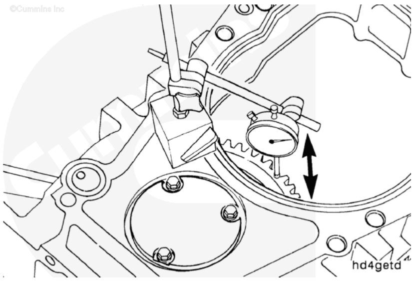


Use a dial indicator.

Measure the idler gear end clearance.

mm		in
0.10	MIN	0.004
0.51	MAX	0.020

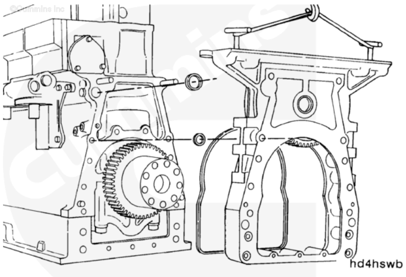
Note : If the end clearance is not within specifications, the thrust bearings must be replaced.



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : If an SAE 1 flywheel housing option is used, there must be two [5/8-11 x 6 1/2 inch] studs, Part Number 3065777, installed in the upper holes of the cylinder block.

Apply a small amount of Lubriplate® 105, multi-purpose lubricant, or gasket adhesive on the seal ring groove, the capscrew counterbores, and the dowel counterbores on the block side of the lower housing.

Install the new rectangular seal ring, with the joint at the top into the groove in the lower housing.

Install the ten new capscrew seals into the capscrew counterbores in the lower housing.

Use Lubriplate® 105, multi-purpose lubricant, or equivalent, on the main rifle seal.

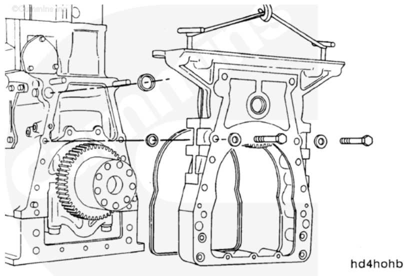
Install the main rifle seal into the counterbore surrounding the main rifle drilling.

Install the lower housing of the rear gear train onto the dowel pins.

Check the alignment of all capscrews seals, rectangular seal, and the main oil rifle seal.

Use two 5/8-11x5 inch capscrews with flat washers in the locations shown.

Tighten the capscrews alternately to pull the lower housing to the block.



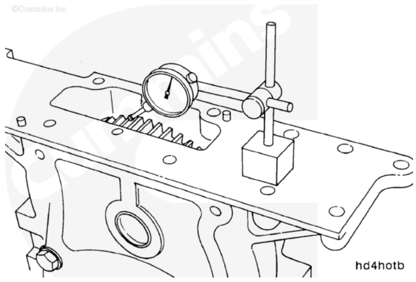
Use a dial indicator. Be sure the two capscrews are holding the lower housing firmly against the cylinder block. Check the gear lash.

mm		in
0.05	MIN	0.002
0.51	MAX	0.015

If the gear lash is above acceptable limits, the rear gear train lower idler gear or crankshaft gear **must** be replaced.

Note : Replace the idler gear first.

If the gear lash is below acceptable limits (or if the gear replacement does **not** correct the lash), replace the housing.



Finishing Steps

- Install the flywheel housing. Refer to Procedure 016-006 (/qs3/pubsys2/xml/en/procedures/20/20-016-006-tr.html).
- Install the oil pan adapter cover plate. Refer to Procedure 007-026 (/qs3/pubsys2/xml/en/procedures/20/20-007-026-tr.html).
- Install the crankshaft rear seal. Refer to Procedure 001-024 (/qs3/pubsys2/xml/en/procedures/20/20-001-024-tr.html).
- Install the flywheel. Refer to Procedure 016-005 (/qs3/pubsys2/xml/en/procedures/20/20-016-005-tr.html).
- Install the upper rear gear drive assembly. Refer to Procedure 009-024 (/qs3/pubsys2/xml/en/procedures/20/20-009-024-tr.html).

- Install the transmission, clutch, and all related components. Refer to the equipment manufacturer's instructions.
- Install the starter. Refer to Procedure 013-020 (/qs3/pubsys2/xml/en/procedures/20/20-013-020-tr.html).
- Fill the engine with oil to the correct level. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/20/20-007-037.html).
- Operate the engine and check for leaks.

Last Modified: 30-Jun-2006